Contents

1 字串

1.1 字典樹 trie

```
struct trie{
    trie *nxt[26]={};
    int cnt=0;
    int sz=0;
};
trie *root=new trie();
void insert(const string &s){
    trie *now=root;
    for(auto i:s){
        if(now->nxt[i-'a']==NULL){
            now->nxt[i-'a']=new trie();
        }
        now=now->nxt[i-'a'];
}}
int qry(string &s){
    trie *now=root;
    for(auto i:s){
        if(now->nxt[i-'a']==NULL) return;
        now=now->nxt[i-'a'];
    }
    return now->cnt;
}
```

1.2 KMP

```
vector<int> kmp(string s,string t){
    if(t.size()>s.size()) return {};
    vector<int> pi(t.size(),0);
    for(int i=1,j=0;i<t.size();i++){
        while(j>0 && t[j]!=t[i]){
            j=pi[j-1];
        }
        if(t[j]==t[i]) j++;
        pi[i]=j;
    }
    vector<int> ans;
    for(int i=0,j=0;i<s.size();i++){
        while(j>0 && t[j]!=s[i]){
            j=pi[j-1];
        }
        if(t[j]==s[i]) j++;

        if(j==t.size()){
            ans.push_back(i-j+1);
            j=pi[j-1];
    }}
    return ans;
}
```

2 最短路徑

2.1 basic

```
// c++ code
#include <bits/stdc++.h>
using namespace std;

int main() {
    // test comment
    cout << "test string\n";
}
```

3 圖論

3.1 凸包問題

```
struct node{
    int x,y;
    bool operator<(const node &other) const{
        return
        (x<other.x) || (x==other.x && y<other.y);
    };
};
struct ConvexHull{
    vector<node> vec;
    int n;
    void init(vector<node> &v){
        vec=v;
        n=v.size();
    }
    int cross(node a,node o,node b){
        return
        (a.x-o.x)*(b.y-o.y)-(b.x-o.x)*(a.y-o.y);
    }
    vector<int> point;
    vector<int> area;
    void build(){
        point=vector<int>(n+1);
        area=vector<int>(n+1);
        vector<node> up,down;
        int upsum=0;
        int downsum=0;
        for(int i=0;i<n;i++){
            while(up.size()>=2 &&
            cross(up[up.size()-2],
                  up[up.size()-1],
                  vec[i])<=0){
                upsum-=cross(up[up.size()-2],
                             {0,0},up[up.size()-1]);
                up.pop_back();

            }
            up.push_back(vec[i]);
            if(up.size()>1)
                upsum+=cross(up[up.size()-2],
                             {0,0},
                             up[up.size()-1]);

            while(down.size()>=2 &&
            cross(down[down.size()-2],
                  down[down.size()-1],
                  vec[i])>=0){
                downsum-=cross(down[down.size()-2],
                               {0,0},down[down.size()-1]);
                down.pop_back();
            }
            down.push_back(vec[i]);
            if(down.size()>1)
                downsum+=cross(down[down.size()-2],
                               {0,0},down[down.size()-1]);

            area[i+1]=abs(upsum-downsum);
```

```cpp

            if(i+1==1) point[i+1]=1;
            else point[i+1]=up.size()+down.size()-2;
    }}
    double getarea(int p){
        return area[p]/2.0;
    }
    int getnum(int p){
        return point[p];
    }
};
```

4 數學

4.1 公式

- 中文測試
- $\sum\limits_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$